

Amphiphilic Marine Antifouling Coatings Based on Zwitterion-Modified Silicone Polymers

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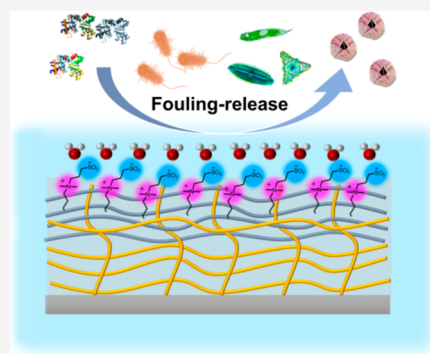


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ABSTRACT: Silicone coatings are widely employed in marine antifouling applications due to their low surface energy. However, in static marine environments, pure silicone coatings are ineffective in preventing the adhesion of marine biofilms, which consist of proteins, marine bacteria, and extracellular matrices, ultimately promoting the attachment of macrofouling organisms. To address the limitations in antifouling performance under static conditions, this study introduces a silicone-based antifouling coating modified with zwitterionic polymers. Sulfobetaine (SB) zwitterionic segments were grafted onto the side chains of poly(dimethylsiloxane) (PDMS) to synthesize the amphiphilic polymer P(DMS-SB), which was incorporated into the PDMS network to create an interpenetrating network-structured silicone coating. The zwitterionic segments effectively inhibited the adhesion of proteins, bacteria, and algae through hydration effects. Compared to pure PDMS coatings, the adhesion of proteins, bacteria, and algae was reduced by 88%, 98.9%, and 99.3%, respectively. Additionally, the coating demonstrated excellent fouling-release properties, achieving a 91.3% removal rate for settled algae under water flow conditions and reducing the simulated barnacle adhesion strength by 68.4%. This coating presents a promising antifouling solution for ships, offshore structures, and aquaculture facilities in static marine environments with significant potential for widespread application.



INTRODUCTION

Marine biofouling, resulting from the accumulation and colonization of marine organisms such as algae, bacteria, and mollusks on submerged surfaces, presents significant economic and environmental challenges.^{1–3} For example, biofouling on ship hulls not only increases weight and surface roughness but also leads to greater fuel consumption and higher greenhouse gas emissions.⁴ Additionally, the introduction of non-native marine species into foreign ecosystems can cause severe ecological disruption.⁵ Marine biofouling progression consists of four stages, namely, base film adhesion (e.g., proteins, polysaccharides, and other organic macromolecules), primary adhesion (e.g., bacteria forming biofilms), secondary adhesion (e.g., diatoms, microalgae spores, and other multicellular organisms), and tertiary adhesion (e.g., barnacles, mussels, and other macrofouling organisms).^{6,7} Effective antifouling can be achieved by inhibiting the adhesion of fouling organisms, particularly the base films and biofilms, on the surfaces of hulls and marine equipment.

Currently, marine antifouling coatings are widely regarded as one of the most convenient, effective, and cost-efficient solutions to address biofouling.^{8,9} However, most commercial antifouling coatings are associated with the release of biocides, including copper oxide, organocopper compounds, and organic

fungicides or herbicides, which pose threats to marine life and the broader marine ecosystem.^{10–13} As environmental concerns intensify, the development of nontoxic and eco-friendly antifouling technologies has become a critical priority.^{14–16}

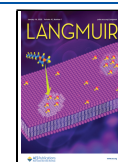
PDMS, known for its high flexibility, low elastic modulus, and low surface energy, significantly reduces the adhesion of hard marine fouling organisms, such as barnacles, and is widely used in fouling-release coatings.^{17–20} The fouling-release performance of the PDMS depends on the high-speed movement of vessels. However, for stationary vessels, PDMS exhibits poor antifouling properties as it cannot prevent the adhesion of marine biofilms composed of proteins and diatoms.^{21,22} To improve the antifouling performance of PDMS, researchers have incorporated hydrophilic polymers, such as PEG, into PDMS through physical blending or chemical modification, producing amphiphilic antifouling

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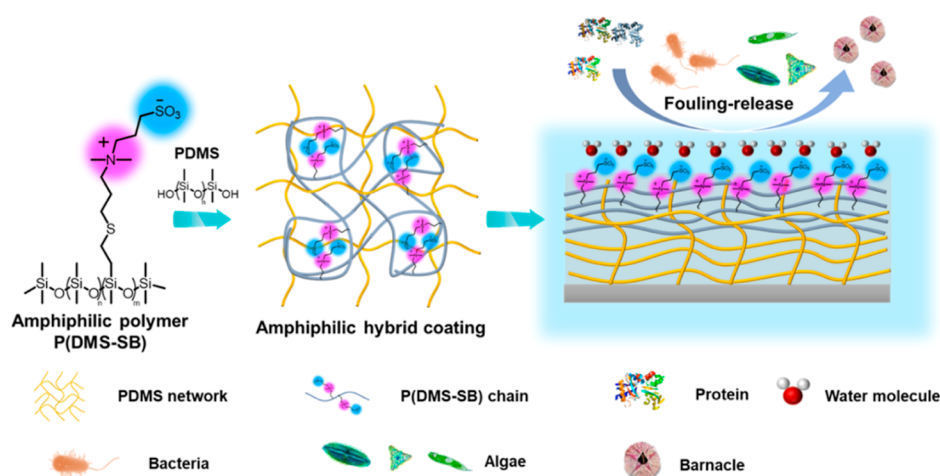


Figure 1. Schematic illustration for the design of the amphiphilic antifouling coatings based on zwitterion-modified silicone.

coatings. The hydrophobic, low-surface-energy PDMS facilitates the easy detachment of marine organisms under hydrodynamic forces, while the hydrophilic polymer binds water molecules tightly, forming a hydration layer that resists biofouling.^{23–25} However, studies have shown that PEG is prone to oxidation and degradation in marine environments, significantly reducing the coating's antifouling efficacy and shortening its lifespan.^{26,27} Zwitterionic materials, which carry an equal number of positive and negative charges, form a stable hydration layer on the surface, effectively inhibiting the adhesion of proteins and microorganisms, thus exhibiting remarkable and long-lasting static antifouling performance compared to PEG.^{28–33} Additionally, zwitterionic materials are nontoxic and environmentally friendly, minimizing adverse environmental impacts, making them widely applicable in antifouling coatings.^{34,35} Despite this, the chemical incompatibility between hydrophilic zwitterionic materials and hydrophobic PDMS leads to microphase separation and surface roughness during fabrication, which degrades the mechanical properties and weakens the coating's antifouling performance.^{28,36,37}

To address the challenges associated with the development and application of marine antifouling coatings, this study utilized an in situ grafting technique to introduce zwitterionic sulfobetaine (SB) groups into the side chains of PDMS, thereby synthesizing the amphiphilic silicone polymer P(DMS-SB), characterized by ultrahydrophilic SB moieties. P(DMS-SB) was then incorporated into a PDMS matrix to create a silicone-based antifouling coating with an interpenetrating polymer network (Figure 1). Laboratory tests confirmed that the coating demonstrated outstanding antifouling and fouling-release properties. This research offers a novel preparation method and antifouling strategy for safeguarding marine equipment and infrastructure in static marine environments.

EXPERIMENTAL SECTION

Materials. Dihydroxyl-terminated polydimethylsiloxane (HO-PDMS-OH, $M_w = 30,000$ g mol⁻¹, from Heowns), methylsilanetriyl triacetate (METES, 96%, from Tianjin Heowns Biochem), vinylmethylsiloxane–dimethylsiloxane copolymer, trimethylsiloxy-terminated [P(DMS-VMS), 800–1200 cSt, $M_w = 28,000$ g mol⁻¹, from Gelest], dibutyltin dilaurate (DBTDL, 98%, from Adamas), 3-(dimethylamino)-1-propanethiol (DPT, 97%, from Beijing Ouhe Technology Co. Ltd.), 1,3-propyl sultone (1,3-PS, 99%, from Shanghai Macklin Biochemical Co., Ltd.), and benzoin dimethyl

ether (DMPA, 99%, from Nanjing Sheng Bicheng Chemical Technology Co. Ltd.) were all purchased without further purification. Solvents tetrahydrofuran (THF, AR) and dichloromethane (CH₂Cl₂, AR) were purchased from Tianjin Kermel Chemical Reagent Co. Ltd. and Concord Technology, respectively.

Synthesis of the Amphiphilic P(DMS-SB). As shown in Figure S1, P(DMS-DMA) was first synthesized. P(DMS-VMS) (5.00 g, 0.18 mmol), DPT (1.00 g, 8.39 mmol), and DMPA (0.03 g) were added to dried THF (15 mL) and placed in a sealed 50 mL flask. The reaction mixture was degassed with argon and irradiated with UV light (365 nm) for 1 h at room temperature. THF and excess DPT were then removed by vacuum drying at 50 °C. The resulting viscous yellow liquid was P(DMS-DMA). The obtained P(DMS-DMA) (5.96 g, 0.18 mmol) was dissolved in dried THF (15 mL), placed in a 50 mL flask, and then cooled to 0 °C. Under vigorous stirring, a solution of 1,3-PS (1.00 g, 8.19 mmol) in THF (5 mL) was added dropwise. After reacting at room temperature for 24 h, the product was collected and washed three times with dried acetone and ether, respectively. The resulting transparent solid was vacuum-dried at room temperature to yield P(DMS-SB).

Preparation of Amphiphilic Silicone Coatings. The preparation method of the amphiphilic silicone coatings is shown in Figure S2. Hydroxyl-terminated polydimethylsiloxane (HO-PDMS-OH, 5.0 g) was dissolved in 15 mL of dichloromethane and stirred for 0.5 h. Methyl triacetoxysilane (METES, 0.5 g), dibutyltin dilaurate (DBTDL, 50 mg) as the catalyst, and varying proportions of the amphiphilic polymer P(DMS-SB) (5, 10, and 15 wt %) dissolved in 15 mL of dichloromethane were then added. The mixture was stirred continuously at room temperature for 1 h, after which it was applied to glass slides and allowed to dry at room temperature for 12 h. The resulting amphiphilic dual-network coatings were named PDMS-SB-X, where X represents the weight percentage of P(DMS-SB) in the coating. Pure PDMS coatings were prepared using the same method. The detailed composition of the coatings is given in Table 1.

Characterization. The ¹H NMR spectra were obtained on a Bruker AV 500 spectrometer (500 MHz) using CDCl₃ as the solvent. The infrared spectra of the amphiphilic polymers and coatings were measured in the wavenumber range of 500–3500 cm⁻¹ by using a

Table 1. Compositions for the Coatings

sample	HO-PDMS-OH (wt %)	METES (wt %)	P(DMS-SB) (wt %)	solvent and DBTDL (wt %)
PDMS	50	5	0	45.0
PDMS-SB-5	50	5	2.5	42.5
PDMS-SB-10	50	5	5.0	40.0
PDMS-SB-15	50	5	7.5	37.5

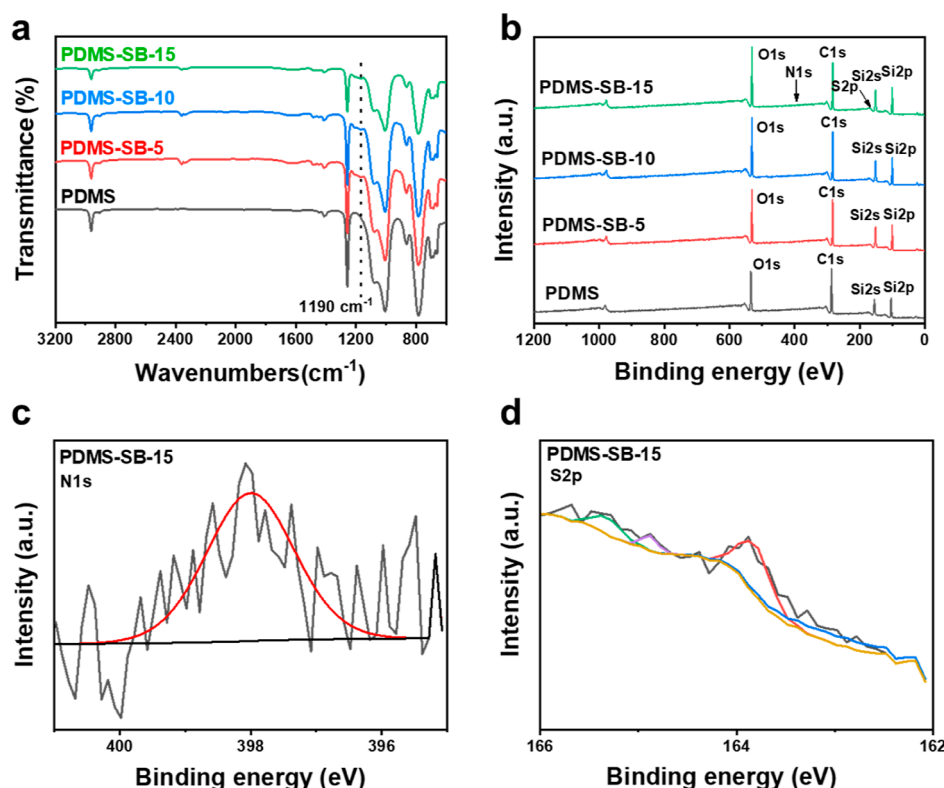


Figure 2. (a) FTIR spectra of different coatings, (b) wide-region XPS spectra of different coatings, (c) N 1s XPS spectra of the PDMS-SB-15 coating, and (d) S 2p XPS spectra of the PDMS-SB-15 coating.

Nicolet 6700 Fourier-transform infrared (FTIR) spectrometer. Scanning electron microscopy (SEM) images were acquired using a SEM device (FEI, Apreo S LoVac, America) equipped with an energy-dispersive spectrometer, operating with a 10 keV electron beam and a working distance of 10 mm, to observe and analyze the elemental content of the coating cross sections. The cross-sectional samples were obtained by brittle breaking in a $-196\text{ }^{\circ}\text{C}$ liquid nitrogen environment, followed by sputter-coating with Pt for SEM examination. X-ray photoelectron spectroscopy (XPS) analysis was conducted using a Thermo Scientific K-Alpha instrument. The glass transition temperature (T_g) of the material was measured by using a differential scanning calorimeter.

Mechanical Property Tests. The mechanical properties of the coatings were evaluated through tensile testing by using a micro-computer-controlled electronic universal testing machine (WDW-SW, China). Samples were prepared and tested according to the ASTM 638 standard by casting the coating into polytetrafluoroethylene molds, producing specimens with dimensions of $75 \times 4 \times 2$ mm. The tensile tests were conducted at room temperature with a strain rate of 5 mm min^{-1} .

Dynamic Contact Angle Tests. To investigate the hydrophilicity/hydrophobicity of the coatings, water contact angles (WCA) were measured at room temperature ($25\text{ }^{\circ}\text{C}$) in air using an optical contact angle testing system (JC2000D1, Shanghai Zhongchen Digital Technology Instruments Co., Ltd.). A small water droplet ($5\text{ }\mu\text{L}$) was placed on the coating surface, and the contact angle was recorded every 15 s over a measurement period of 4 min. The droplet was sufficiently small for gravitational effects to be negligible.

Static Anti-fouling Adhesion Performance Tests. *Anti-protein Adhesion Test.* PDMS-SB-X and control PDMS samples were incubated in FITC-BSA PBS solution (0.5 mg mL^{-1}) for 2 h under dark conditions. The sample surfaces were then gently rinsed with PBS. Adhesion of FITC-BSA to the coatings was observed using an inverted fluorescence microscope (Nikon Eclipse Ti-S, Japan) under blue-light excitation. Additionally, the amount of BSA adhered to the coatings was quantified via ELISA. The coatings were incubated

in BSA solution (0.5 mg mL^{-1}) at $37\text{ }^{\circ}\text{C}$ for 2 h, followed by rinsing with PBS. The presence of protein in the samples was confirmed using a standard ELISA protocol.

Anti-bacterial Adhesion Test. Antibacterial tests were performed using *Escherichia coli* (*E. coli*, Gram-negative, G^-) and *Staphylococcus aureus* (*S. aureus*, Gram-positive, G^+). The PDMS-SB-X coatings on glass substrates ($1.0 \times 1.0\text{ cm}^2$) were sterilized using 75 wt % ethanol and UV light and then immersed in bacterial suspensions ($1 \times 10^6\text{ CFU mL}^{-1}$). After incubation at $37\text{ }^{\circ}\text{C}$ for 4 h, the coatings were washed three times with 1 mL of PBS to remove adherent bacteria. From the final wash, $100\text{ }\mu\text{L}$ was spread onto LB agar plates and the colony-forming units (CFU) of viable bacteria in the wash were counted. Additionally, GFP-expressing *E. coli* (GFP-*E. coli*) was used to assess the resistance of the coatings to bacterial adhesion. Sterilized samples were immersed in bacterial suspensions ($1 \times 10^6\text{ CFU mL}^{-1}$) and incubated at $37\text{ }^{\circ}\text{C}$ while being shaken at 150 rpm for 4 h. The samples were then gently washed three times with PBS to remove unattached bacteria. Fluorescent *E. coli* adhered to the coating surface were visualized using an inverted fluorescence microscope (Nikon Eclipse Ti-S) under blue light excitation.

Anti-diatom Adhesion Test. Following previous reports, the marine diatom *Phaeodactylum tricornutum* was cultured in sterilized artificial seawater (ASW). The coatings applied on glass substrates were immersed in ASW for 3 days and then transferred to 24-well plates. A 1 mL suspension of diatoms ($1 \times 10^6\text{ cells mL}^{-1}$) was added to each well and incubated at $25\text{ }^{\circ}\text{C}$ in a climate chamber with a 12 h light/dark cycle. After 24 h, the coating surfaces were gently rinsed with sterilized ASW to remove loosely attached diatom cells. Fluorescent diatom cells adhered to the coatings were then observed using an inverted fluorescence microscope under blue light excitation. The number of *P. tricornutum* cells attached to the coatings was counted in 30 fields of view per replicate, and the average number of attached diatoms was calculated, providing statistical data with a 95% confidence interval.

Dynamic Fouling-Release Performance Tests. *Adhered Diatom Flow Scouring Tests.* First, a static diatom attachment test

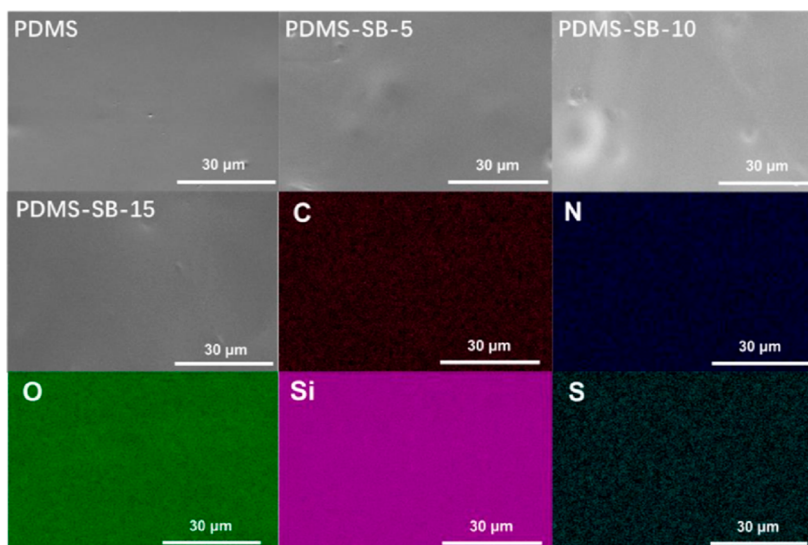


Figure 3. Representative cross-sectional SEM images of different coatings and elemental mapping images of the PDMS-SB-15 coating.

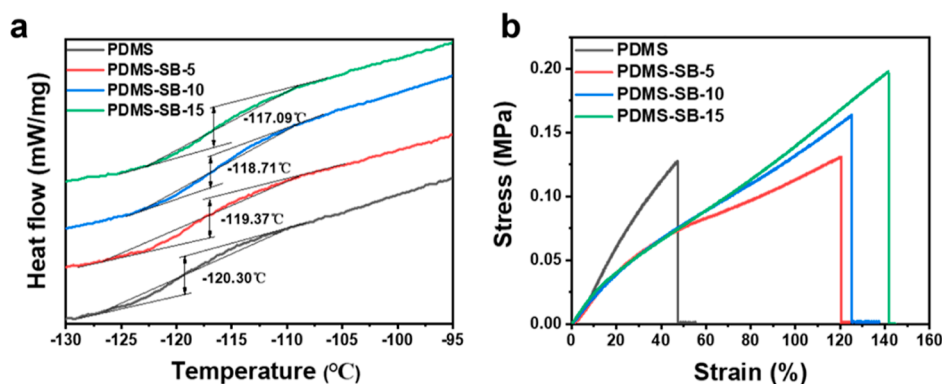


Figure 4. (a) DSC curves of the PDMS and PDMS-SB-X coatings and (b) tensile stress–strain curves of the PDMS and PDMS-SB-X coatings.

was conducted on the sample surface. Afterward, the samples were subjected to water flow at a velocity of 2.87 m/s (approximately 5.6 knots) for 5 s to assess the removal rate of *P. tricornutum*. The samples were observed under an inverted fluorescence microscope before and after the flow experiment, and the fouling removal rate (R) was calculated using the formula: $R = (N_0 - N_1)/N_0 \times 100\%$, where N_0 represents the number of algae attached to the sample surface before the flow test, and N_1 represents the number of algae remaining after the test.

Mimetic Barnacle Detachment Tests. To evaluate the coatings' resistance to barnacle adhesion, the shear adhesion strength of mimetic barnacles was measured according to ASTM D 5618-94 (1994). A PosiTest device (DeFelsko) was used for the simulated barnacle tests. The barnacle models were adhered to the coating surface by using epoxy glue. A force parallel to the coating surface was then applied at a rate of 0.1 mm min⁻¹ until the simulated barnacles detached from the surface.

RESULTS AND DISCUSSION

Chemical Structure and Microscopic Morphology of the Coatings. The structures of P(DMS-DMA) and P(DMS-SB) were characterized by ¹H NMR spectroscopy. The P(DMS-DMA) ¹H NMR spectrum is shown in Figure S3: ¹H NMR (500 MHz, CDCl₃): δ ppm 2.54–2.63 (ddd, 4H), 2.34–2.46 (d, 2H), 2.22–2.33 (s, 6H), 1.71–1.85 (m, 2H), 0.83–0.95 (m, 2H), 0.08–0.12 (d). The ¹H NMR spectrum of P(DMS-SB) is shown in Figure S4: ¹H NMR (500 MHz,

CDCl₃): δ ppm 3.72–3.83 (tdd, 2H), 3.12–3.41 (m, 6H), 2.55–2.60 (d, 4H), 1.71–1.85 (m, 2H), 1.83–1.90 (s, 2H), 0.05–0.12 (d). The results confirmed the successful synthesis of zwitterion-modified silicone polymer P(DMS-SB).

The functional groups on the coating surface were analyzed by using FTIR spectroscopy. As shown in Figure 2a, for the PDMS sample, the stretching vibration peaks at 2963 and 1010–1120 cm⁻¹ correspond to the –CH₃ and Si–O–Si groups in PDMS, respectively, while the peak observed at 802 cm⁻¹ is attributed to Si–C bending vibrations. After the incorporation of the P(DMS-SB) polymer, new stretching vibration peaks appear at 1190 and 921 cm⁻¹, corresponding to –SO₃⁻ and –R₄N⁺, respectively. This indicates the successful grafting of the zwitterionic siloxane polymer into the PDMS coating. Further analysis of the elemental composition of the coating surface was performed using XPS. As shown in Figure 2b, the survey spectrum reveals characteristic peaks of C 1s, O 1s, Si 2s, and Si 2p, corresponding to the PDMS matrix. Additionally, the high-resolution XPS spectrum of the PDMS-SB-15 sample displays characteristic N 1s and S 2p peaks, confirming the presence of the SB groups on the coating and demonstrating that P(DMS-SB) has been successfully integrated into the PDMS matrix (Figure 2c,d).

The incorporation of the amphiphilic polymer P(DMS-SB) into the hydrophobic PDMS matrix raises the concern of potential phase separation due to the disparity in material

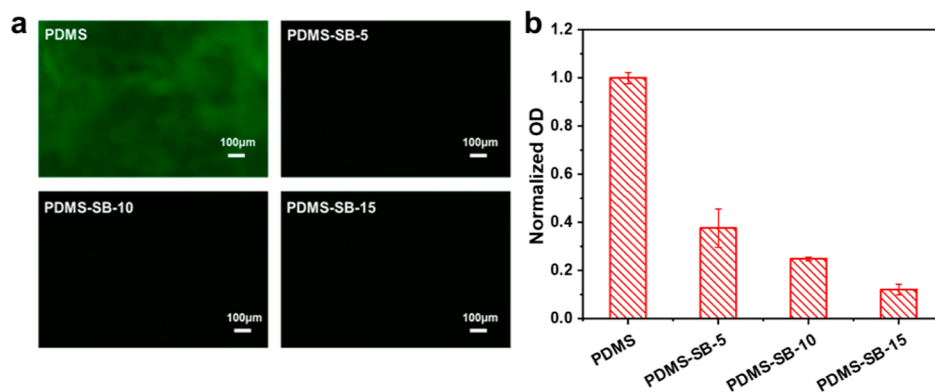


Figure 5. (a) Representative fluorescence microscopy images of attached BSA-FITC (green) from a 0.5 mg mL⁻¹ solution after exposure to different coatings for 2 h in the dark; (b) amount of BSA attached on different coatings tested by ELISA.

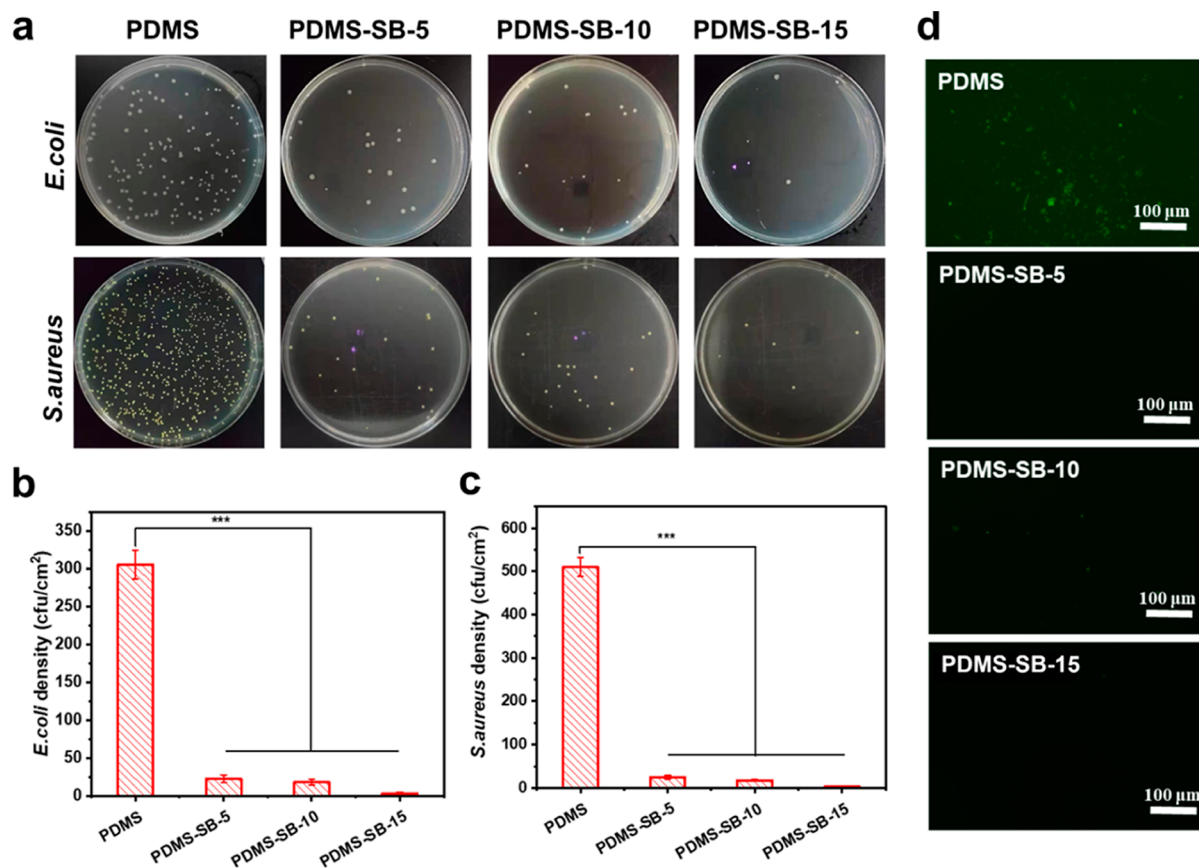


Figure 6. (a) Photographs of *E. coli* and *S. aureus* coated on LB medium after incubation with different coatings for 4 h and the number of (b) *E. coli* and (c) *S. aureus* attached on the coatings. (d) Representative fluorescence microscopy images of attached GFP-*E. coli* from a 1×10^6 CFU mL⁻¹ suspension after exposure to the coatings for 4 h.

polarity, which could significantly impact the coating's performance. The cross-sectional morphology of the coatings was characterized via SEM (Figure 3), and the images reveal that both the PDMS and PDMS-SB coatings exhibit uniform cross sections with no evident signs of phase separation. This indicates that P(DMS-SB) is well integrated into the PDMS network. Furthermore, the EDS analysis demonstrates a homogeneous distribution of nitrogen (N) and sulfur (S) elements throughout the coating, confirming that P(DMS-SB) is evenly dispersed within the entire coating system. This uniform dispersion effectively prevents phase separation, thereby preserving the coating's performance.

Mechanical and Thermal Properties of the Coating.

The glass transition temperature (T_g) and mechanical properties were investigated by DSC and tensile testing. From the DSC curves of the coatings, it can be observed that PDMS-SB-X and PDMS exhibit similar T_g values, specifically -119.37 , -118.71 , and -117.09 °C, respectively (Figure 4a). The T_g of PDMS-SB-15 is 3.2 °C higher than that of PDMS, indicating a reduction in the mobility of the polymer chains within the coating material. This decrease in chain mobility is attributed to the incorporation of P(DMS-SB) into the PDMS network, forming an additional network through electrostatic interactions between zwitterions.^{38,39} The presence of this

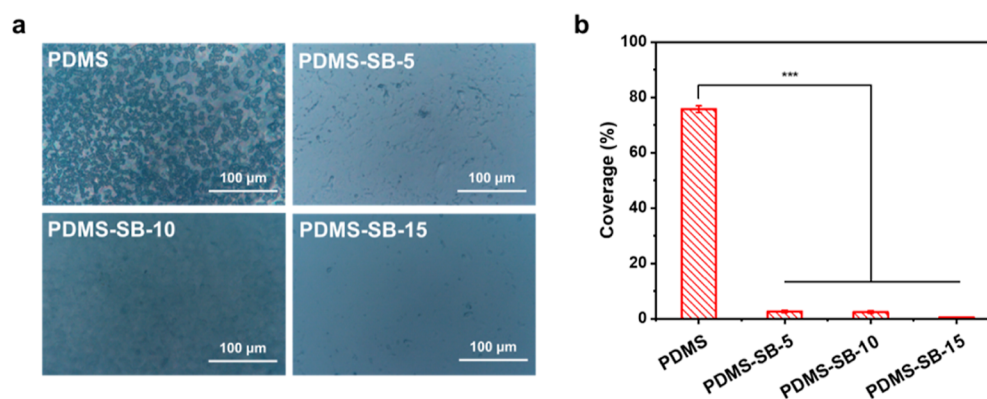


Figure 7. (a) Representative optical microscopy images of attached *P. tricornutum* from a 1×10^6 cell mL^{-1} suspension after exposure to the coatings for 24 h. (b) Coverage of BSA *P. tricornutum* attached on different coatings.

secondary network hinders the movement of the original PDMS network segments.³³ Moreover, the coatings incorporating the PDSB polymer display only a single glass transition temperature, indicating the absence of significant phase separation between PDMS and the amphiphilic polymer PDSB.¹⁹ This is likely due to the formation of a homogeneous and stable interpenetrating network between PDSB and the PDMS matrix.

Additionally, mechanical performance tests were conducted on the coatings, and the tensile stress–strain curves are presented in Figure 4b. The tensile strength and elongation at break of PDMS were measured to be 0.13 MPa and 47.5%, respectively. Upon incorporation of P(DMS-SB), the tensile strength of the coatings increased, with PDMS-SB-15 exhibiting a 54.9% enhancement in tensile strength and a 198.5% improvement in elongation at break compared to PDMS. This enhancement in both strength and flexibility can likely be attributed to the presence of P(DMS-SB), which reinforces the coating. The formation of an interpenetrating double network may increase the cross-linking density of the coating, thereby improving the coating's mechanical strength. Furthermore, as depicted in the figure, during the tensile process, the PDMS network primarily resists deformation until a strain of 47.5% is reached. Beyond this point, the electrostatic interactions between the SB groups on the P(DMS-SB) chains become the dominant source of tensile resistance. This further substantiates the formation of a dense interpenetrating network.

Static Anti-fouling Adhesion Performance. The time a ship spends docked at port constitutes a significant portion of its overall operational time during which the ship remains in a nearly static state. Therefore, it is crucial to study the static antifouling performance of coatings.^{40–42}

Anti-protein Adhesion Performance. Antiprotein adhesion performance is a key parameter for evaluating the antifouling properties of coatings as the initial step in marine biofouling is the attachment of a protein-based conditioning layer.^{43–46} Therefore, this study utilized the representative fluorescent protein FITC-BSA to assess the antiprotein adhesion performance of the coatings. As shown in Figure 5a, almost no fluorescent protein adhered to the PDMS-SB coating, while the PDMS surface was nearly fully covered with fluorescent BSA. In addition, protein adsorption was evaluated using the ELISA test, and compared to PDMS, the protein adhesion on the surface of PDMS-SB-15 was reduced by 88.0%. These results indicate that the hydrophilic SB groups

effectively reduce the level of protein adsorption on the coating (Figure 5b). This is attributed to the zwitterionic SB groups being electrically neutral, which minimizes electrostatic interactions between proteins and the coating, and their ability to interact with water molecules to form a stable hydration layer, thus providing excellent antiprotein properties.

Anti-bacterial Adhesion Performance. Antibacterial adhesion performance is a critical indicator for evaluating the antifouling effectiveness of coatings. In marine environments, a large number of bacteria act as primary microfouling organisms, attaching early to underwater substrates and providing rich nutrients and favorable conditions for the subsequent attachment of fouling organisms.^{47,48} *E. coli*, a common Gram-negative bacterium, and *S. aureus*, a common Gram-positive bacterium, have been widely used to assess the antibacterial adhesion properties of antifouling coatings. As shown in Figure 6a–c, after coculturing different samples with bacteria for 4 h, bacterial adhesion on the PDMS-SB-15 coating surface was reduced by 98.9% compared to the PDMS coating. Furthermore, the fluorescent *E. coli* adhesion experiment shown in Figure 6d indicates that almost no fluorescent *E. coli* was observed on the surface of the PDMS-SB coating, while a significant amount of fluorescent *E. coli* adhered to the PDMS coating. These results demonstrate that the novel amphiphilic antifouling coating exhibits excellent antibacterial adhesion performance. The introduction of hydrophilic zwitterionic SB groups significantly improves the antibacterial adhesion capacity of PDMS coatings as the electrostatic interactions of the zwitterionic SB groups induce the formation of a hydrated barrier, protecting the coating surface from bacterial adhesion and contamination.

Anti-diatom Adhesion Performance. Diatoms, as an important early-stage fouling organism, facilitate the colonization and growth of other microorganisms by providing favorable conditions.^{49,50} Therefore, this study focused on the attachment of the representative diatom *P. tricornutum* on coating surfaces. Figure 7 presents the diatom adhesion results after 24 h of cocultivation with the coatings. The algal coverage areas for PDMS, PDMS-SB-5, PDMS-SB-10, and PDMS-SB-15 were 75.7%, 2.6%, 2.4%, and 0.5%, respectively. The number of algae adhering to the surface of the PDMS-SB-15 coating was reduced by 99.3% compared to the PDMS coating, indicating that the introduction of zwitterionic SB groups significantly enhances the antifouling performance of the coating against algal adhesion.

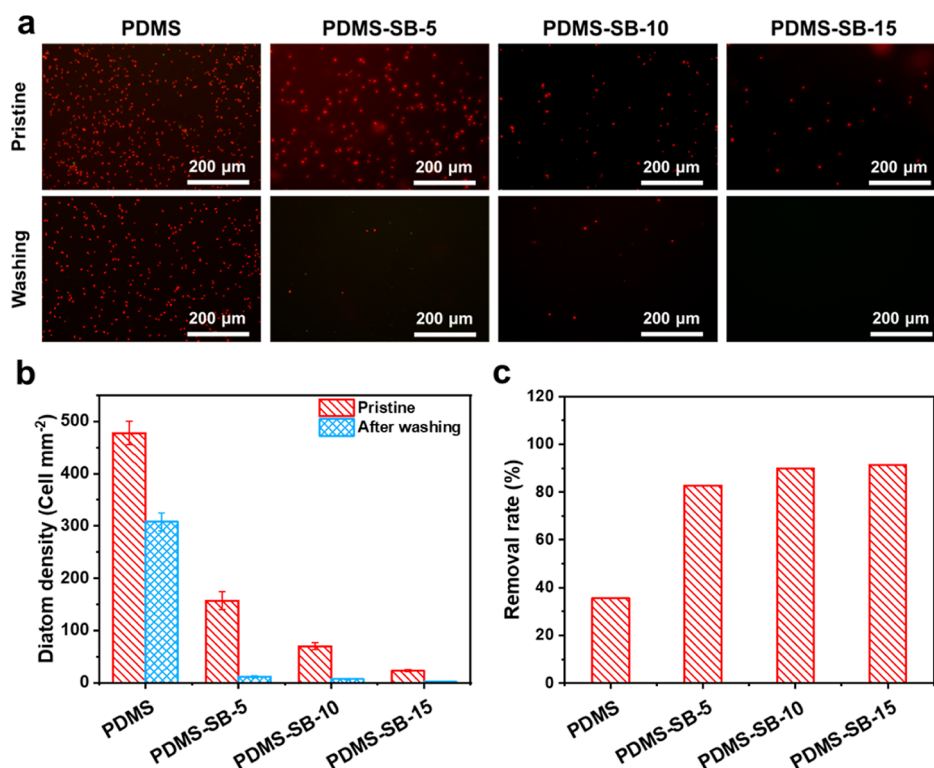


Figure 8. (a) Representative optical microscopy images of the coatings before and after washing; (b) number of *P. tricornutum* attached to the coatings before and after washing; (c) removal rate of *P. tricornutum* on the coating surface.

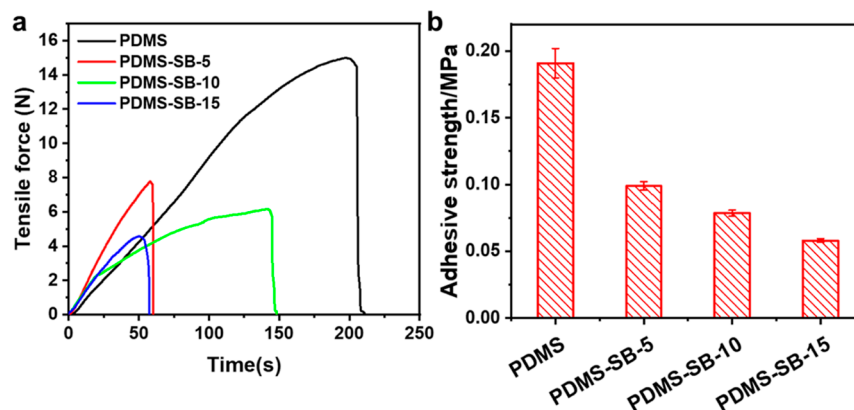


Figure 9. (a) Tensile stress curves of pseudo-barnacles adhering to the coatings. (b) The adhesive strength of pseudo-barnacles.

Dynamic Fouling-Release Performance. During a ship's voyage, antifouling coatings with excellent fouling-release performance can significantly reduce the accumulation of fouling organisms on the vessel's surface. To investigate the dynamic fouling-release performance of the coatings, algae removal flow tests and barnacle adhesion tests were conducted.^{51,52}

When a ship docks at a port for a short period, diatoms and other organisms can adhere to the hull. Therefore, a simulated algae removal flow test was conducted. As shown in Figure 8, the removal rate of *P. tricornutum* from the surface of the PDMS coating was only 35.6%. However, the introduction of hydrophilic zwitterionic SB groups increased the algae removal rate by more than 47.2%, with the PDMS-SB-15 coating achieving a removal rate of 91.3% for surface-attached *P. tricornutum*, demonstrating an excellent fouling-release performance. This is because the zwitterionic SB segments

migrating to the coating surface in seawater interact strongly with water molecules, forming a stable hydration layer. This hydration layer effectively facilitates the dynamic removal of fouling organisms. These results indicate that fouling organisms attached to the hull surface during navigation can be effectively removed by the shear forces of water flow.

Barnacles are notorious marine fouling organisms and are one of the most harmful species to ship hulls. Therefore, a simulated barnacle adhesion test was employed to assess the antibarnacle adhesion capability of the PDMS-SB coatings. The adhesion strength of simulated barnacles was measured according to ASTM D 5618-94 (1994).^{53–55} As shown in Figure 9, the adhesion strength of simulated barnacles on the surface of the PDMS-SB coatings was significantly reduced compared with the PDMS coating. The adhesion strength of simulated barnacles on PDMS-SB-15 decreased by 68.4% (from 0.19 to 0.06 MPa) compared to PDMS. These results

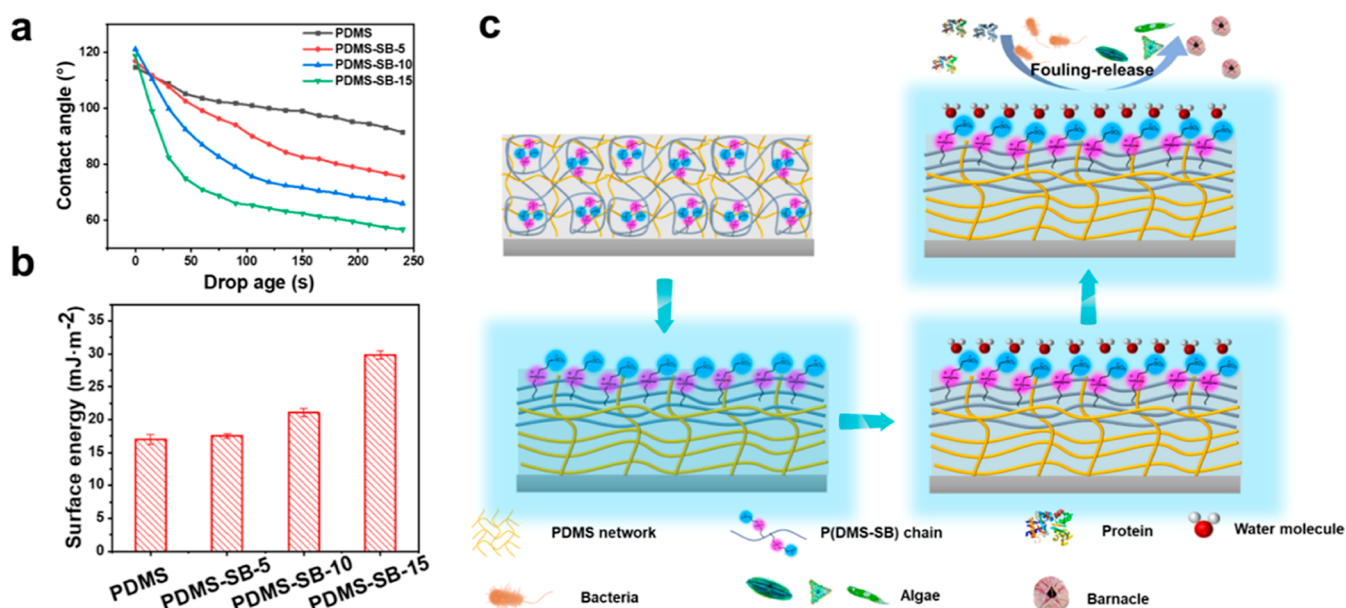


Figure 10. (a) The water dynamic contact angle of different coatings. (b) The surface energy of different coatings. (c) The supposed antifouling mechanism of the amphiphilic coating.

indicate that the lower adhesion strength of the coatings facilitates the easy removal of large fouling organisms such as barnacles under the shear forces of seawater.

Antifouling Mechanism. In summary, the amphiphilic silicone PDMS-SB antifouling coating exhibits excellent antifouling properties. This section explores the antifouling mechanism through studies on the hydrodynamic contact angle and surface energy of the coating. Figure 10a presents the dynamic water contact angle. The variation of the contact angle over time was used to characterize the migration of hydrophilic polymer chains from the interior of the coating to the surface. It can be observed that the introduction of hydrophilic SB groups significantly accelerates the reduction rate of the hydrodynamic contact angle. For PDMS-SB, the hydrodynamic contact angle decreases rapidly within 1 min, after which the rate of decline slows. This indicates that the hydrophilic chains of PDMS-SB quickly migrate to the surface within 1 min, forming a hydration layer.⁵⁶ Furthermore, as shown in Figure 10b, the surface energy of the coating increases with the incorporation of more hydrophilic SB groups. The surface energy of the PDMS-SB-15 coating ranges between 20 and 30 mN/m, which corresponds to the critical surface energy described in the Baier curve for minimal adhesion.^{51,52} Therefore, the antifouling mechanism of the amphiphilic PDMS-SB coating is illustrated in Figure 10c. When the amphiphilic coating is immersed in water, the hydrophilic SB groups migrate to the surface, forming a hydration layer that prevents the adhesion of proteins, bacteria, and algae.^{57–59} At the same time, the coating retains its low surface energy fouling-release properties, enabling the easy removal of fouling organisms during a ship's voyage.

CONCLUSIONS

In this work, a novel zwitterionic amphiphilic polymer, P(DMS-SB), was designed and synthesized, leading to the development of a new silicone-based marine antifouling coating that effectively resists the attachment of fouling organisms, particularly in static marine environments. Based

on the electrostatic interactions of the SB groups on P(DMS-SB), an interpenetrating polymer network was formed with the PDMS matrix, enhancing the compatibility of the hydrophilic zwitterionic groups with the hydrophobic PDMS and improving the mechanical properties of the coating. Through the hydration effect of the zwitterionic groups, the coating effectively inhibited the adhesion of proteins, bacteria, and diatoms, demonstrating an excellent static antifouling performance. Furthermore, simulated algae removal and barnacle adhesion tests confirmed that the material also exhibits outstanding dynamic fouling-release properties. The development of this coating offers a novel and effective antifouling solution for marine equipment and facilities in static marine environments.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.4c04332>.

Synthesis route of amphiphilic polymers, preparation of the amphiphilic silicone coatings, and ¹H NMR spectra of P(DMS-DMA) and P(DMS-SB) (PDF)

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Notes

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